

ANALYSIS OF CONSUMER PERCEPTION AND GROWTH OF PLANT-BASED MEAT AND MILK PRODUCTS

Project report submitted by

SIMRANDEEP KAUR (PLDM-ABM-2151)

in partial fulfillment of the requirements for the award of the degree

Post Graduate Diploma in Management (Agribusiness Management)

under the guidance of

Dr. N.A.Vijay Avinashilingam



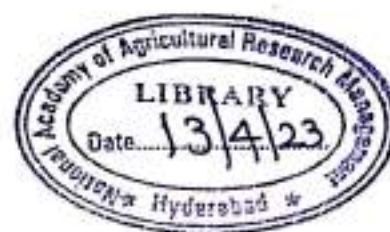
भाकअनुप - राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
ICAR-NATIONAL ACADEMY OF AGRICULTURAL RESEARCH MANAGEMENT
(ISO 9001:2015 CERTIFIED)
RAJENDRANAGAR, HYDERABAD-500030
TELANGANA, INDIA



APRIL 2023

-OR REFERENCE ONLY

Not to be Issued



ANALYSIS OF CONSUMER PERCEPTION AND GROWTH OF PLANT-BASED MEAT AND MILK PRODUCTS

Project report Submitted by

SIMRANDEEP KAUR (PGDM-ABM-2151)

in partial fulfillment of the requirements for the award of the degree

Post Graduate Diploma in Management (Agribusiness Management)

under the guidance of

Dr. N.A.Vijay Avinashilingam

भाकृअनुप — राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
ICAR-NATIONAL ACADEMY OF AGRICULTURAL RESEARCH MANAGEMENT
(ISO 9001:2015 CERTIFIED)
RAJENDRANAGAR, HYDERABAD- 500 030
TELANGANA, INDIA



APRIL 2023

भाकअनुप - राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी

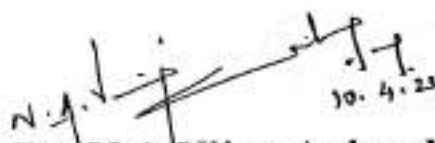
ICAR-NATIONAL ACADEMY OF AGRICULTURAL RESEARCH MANAGEMENT
(ISO 9001:2015 CERTIFIED)

RAJENDRANAGAR, HYDERABAD- 500 030
TELANGANA, INDIA



BONAFIDE CERTIFICATE

Certified that this project report "ANALYSIS OF CONSUMER PERCEPTION AND GROWTH OF PLANT-BASED MEAT AND MILK PRODUCTS" is the bonafide work of "SIMRANDEEP KAUR" who carried out the project work under my guidance.


10. 4. 23

Dr. N.A. Vijay Avinashilingam

Mentor

Principal Scientist & Professor
Extension Systems Management



भाकअनुप - राष्ठीय कृषि अनुसंधान प्रबंध अकादमी

ICAR-NATIONAL ACADEMY OF AGRICULTURAL RESEARCH MANAGEMENT
(ISO 9001:2015 CERTIFIED)

RAJENDRANAGAR, HYDERABAD- 500 030
TELANGANA, INDIA



STUDENT DECLARATION

This is to declare that I, **Simrandeep kaur**, student of Post Graduate Diploma Management in Agri-Business Management (2021-2023) ICAR - NAARM, Hyderabad have given original data, and information, and maintained full confidentiality to the best of my knowledge in the project report titled "**Analysis of consumer perception and growth of plant-based meat and milk products**" and no part of the information has been used for any other assignment but the partial fulfillment of the requirements towards the completion of the said project.

Simrandeep Kaur

PGDM-ABM-2151

ICAR-NAARM

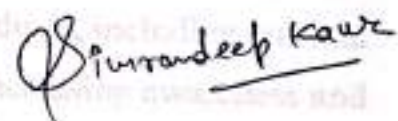
Place: Hyderabad

Date: 10th April, 2023

ACKNOWLEDGEMENT

This acknowledgment is not merely a catalogue of names but an expression of a deep sense of gratitude to all those who helped me in this project directly or indirectly.

I would like to use this opportunity to offer my sincere and heartfelt gratitude to **ICAR-NAARM, Hyderabad** for helping me to set the foundation for reason and pragmatism in my life. These values are somewhat reflected in this study. I want to start by expressing my sincere gratitude, the debt of gratitude, and respect to **Dr. N.A. Vijay Avinashilingam (Principal Scientist, Extension Systems Management)** for his excellent leadership, original, inspiring, and compassionate management of the difficult task, as well as the affection displayed throughout the project. I also like to express my gratitude to **Dr. Yashavanth B S (Scientist, Agricultural Statistics)**, who assisted me in finishing the statistical analysis. Last but not least, I bow before the Almighty and my parents, who have always carried the torch for me throughout my career by guiding me in the proper direction and encouraging me to give my all.



SIMRANDEEP KAUR

PGDM-ABM

ICAR-NAARM

HYDERABAD

EXECUTIVE SUMMARY

The study explores the growth and potential of the plant-based meat and milk industry in India. It examines the factors contributing to the rise of plant-based alternatives, the challenges and opportunities for the industry, and consumer perception and acceptance of plant-based meat and milk in India. The plant-based meat and milk industry in India is still in its early stages, but it is growing rapidly. The market for plant-based meat and milk products is expected to reach \$10 billion.

The Study found that the plant-based meat and milk market in India is on the cusp of significant growth, driven by several factors. First, increasing health concerns related to the consumption of meat and milk products have led consumers to seek healthier and more sustainable alternatives. Second, rising environmental concerns have led to a shift towards plant-based options that have a lower carbon footprint. Third, ethical considerations, such as animal welfare, have led to a growing interest in plant-based meat and milk among consumers.

Some of the most popular plant-based meat products in India are soy-based chicken, plant-based kebabs, and vegan meatballs. For plant-based milk, options like soy milk, almond milk, and oat milk are gaining popularity. While there is significant potential for growth, there are also challenges to the widespread adoption of these products, including cultural and taste preferences, pricing, and distribution. However, with increasing awareness and innovation, the future looks bright for plant-based meat and milk products in India. There are several companies operating in the plant-based meat and milk industry in India. Some of the major players include GoodDot, Vezlay, GreenDot, and Veggie Champ.

However, several challenges exist for India's plant-based meat and milk industry. High production costs are a major challenge, as are limited distribution channels and cultural preferences for meat and dairy-based products. Additionally, there is a lack of awareness among consumers regarding the nutritional value and taste of plant-based meat and milk,

which could impact consumer adoption. The study concludes that the plant-based meat and milk industry in India has immense potential to disrupt the traditional meat and dairy industry.

1.1 Introduction	1
1.2 Objectives of the study	2
1.3 Scope of the study	3
1.4 Limitations of the study	4
1.5 Organization of the study	5
2. Literature Review	6
2.1 Plant-based meat and milk products	6
2.2 Consumer perception and adoption of plant-based meat and milk products	7
2.3 Market trends and opportunities for plant-based meat and milk products in India	8
2.4 Challenges faced by plant-based meat and milk products in India	9
2.5 Government policies and regulations related to plant-based meat and milk products	10
2.6 Future prospects of plant-based meat and milk products in India	11
3. Methodology	12
3.1 Research design	12
3.2 Data collection	13
3.3 Data analysis	14
3.4 Ethical considerations	15
4. Results and Discussion	16
4.1 Consumer perception and adoption of plant-based meat and milk products	16
4.2 Market trends and opportunities for plant-based meat and milk products in India	17
4.3 Challenges faced by plant-based meat and milk products in India	18
4.4 Government policies and regulations related to plant-based meat and milk products	19
4.5 Future prospects of plant-based meat and milk products in India	20

TABLE OF CONTENTS

BONAFIDE CERTIFICATE	ii
STUDENT DECLARATION.....	iii
ACKNOWLEDGEMENT.....	iv
EXECUTIVE SUMMARY	vi
CHAPTER 1- INTRODUCTION	x
1.1 Introduction.....	1
1.2 Statement of the problem	2
1.3 Need for study.....	2
1.4 Objectives	3
1.5 Scope of the study	3
1.6 Limitations of the study	4
CHAPTER 2 - REVIEW OF LITERATURE	5
3.1 Elucidate the trends and current market size of plant-based meat and milk products.....	5
3.2 Study the consumer perception towards plant-based meat and milk products	6
3.3 Identify the factors influencing the growth of plant-based meat and milk products	7
CHAPTER 3 -RESEARCH METHODOLOGY	8
2.1 Selection of study area.....	8
2.2 Location of study area.....	8
2.3 Sampling procedure	8
2.4 Variables and their measurement.....	8
2.5 Research Design.....	8
2.6 Sampling technique.....	8
2.7 Data Collection method	9
2.8 Data Analysis and statistical tools applied.....	9
CHAPTER 4 - RESULTS AND DISCUSSION	11
4.1 Elucidate the trends and current market size of plant-based meat and milk products.....	11
4.1.1 Global Market Overview.....	11
4.1.2 Trends in the global plant-based food market.....	12
4.1.3 Plant-based meat products	14
4.1.4 Plant-based dairy products.....	19
4.2 Consumer perception towards plant-based meat and milk products	23
4.2.1 Location of Respondents.....	24

4.2.2 Occupation of Respondents.....	24
4.2.3 Age of Respondents	25
4.2.4 The annual income of Respondents	25
4.2.5 Educational background of Respondents	26
4.2.6 Awareness of Respondents.....	26
4.2.7 Monthly Expenditure.....	26
4.2.8 Distribution channel of plant-based milk and meat products.....	27
4.2.9 Reason behind choosing plant-based products.....	28
4.2.10 Chi-square test.....	29
4.2.11 Preference for plant-based meat products	31
4.2.12 Preference for plant-based meat brands	31
4.2.12 Source preference for plant-based meat products	32
4.2.13 Preference for plant-based milk products.....	32
4.2.14 Preference for plant-based milk brands.....	33
4.2.15 Source preference for plant-based milk products.....	34
4.3 factors influencing the growth of plant-based meat and milk products.....	34
4.3.1 Factors influencing the growth of plant-based meat	34
4.3.2 Factors influencing the growth of plant-based milk.....	35
CHAPTER 5 – CONCLUSION AND RECOMMENDATIONS	35
5.1 To elucidate the trends and current market size of plant-based meat and milk products ..	36
5.2 To study the consumer perception towards plant-based meat and milk products	36
5.3 To identify the factors influencing the growth of plant-based meat and milk products. ...	36
5.4 Recommendations.....	37
REFERENCES	38
APPENDICES	40

LIST OF FIGURES

Figure 1:Revenue of plant-based meat globally	11
Figure 2:Revenue of plant-based milk globally	12
Figure 3:Revenue of plant-based meat India.....	14
Figure 4:Average revenue per capita of plant-based meat India	15
Figure 5:Volume of plant-based meat India	15
Figure 6:Price per unit of plant-based meat India	16
Figure 7:Revenue of plant-based milk India	19
Figure 8:Revenue of plant-based milk India	20
Figure 9:Average volume per capita of plant-based milk India	20
Figure 10:Price per unit of plant-based milk India.....	21
Figure 11:Location of Respondents.....	24
Figure 12:Occupation of Respondents	24
Figure 13:Age of Respondents	25
Figure 14:Annual Income of Respondents	25
Figure 15:Educational Background of Respondents	26
Figure 16:Awareness of Respondents	26
Figure 17:Monthly Expenditure of Respondents.....	27
Figure 18:Distribution channel.....	27
Figure 19:Reason behind choosing plant-based products	28
Figure 20:Preference for plant-based meat products.....	31
Figure 21:Source preference for plant-based meat products	32
Figure 22:Preference for plant-based milk products	33
Figure 23:Source preference for plant-based milk products	34
Figure 24:Factors influencing the growth of plant-based meat.....	34
Figure 25:Factors influencing the growth of plant-based meat.....	35

LIST OF TABLES

Table 1: Chi-square test	29
Table 2: Preference for plant-based meat brands	31
Table 3: Preference for plant-based milk brands	33

CHAPTER 1- INTRODUCTION

1.1 Introduction

Plant-based meat and plant-based milk are becoming increasingly popular in India, as people are becoming more conscious of their health and the environment. Plant-based meat is made from ingredients such as soy, wheat, pea protein, and other plant-based sources. It is designed to look, taste, and cook like meat, but without the animal products. Plant-based meat is a healthier alternative to traditional meat as it is typically lower in saturated fat and higher in fibre. It is likewise greater environmentally pleasant because it calls for much less land, water, and electricity to produce. Several companies in India are now offering plant-based meat options, including Good Dot, Veggie Champ, and Yumlane. These companies are making plant-based meat products such as burgers, kebabs, and sausages, which are available in grocery stores, online, and some restaurants.

Plant-based milk, also known as non-dairy milk, is made from plant-based sources such as soy, almond, coconut, rice, and oats. It is a popular alternative to cow's milk for those who are lactose intolerant, allergic to milk protein, or following a vegan lifestyle. Plant-based milk is typically lower in calories and fat than cow's milk and can be fortified with vitamins and minerals to provide similar nutritional benefits. Several plant-based milk brands are now available in India, including NutriMoo, Goodmylk, and The Butternut Co. These brands offer a variety of non-dairy milk options, including soy, almond, and oat milk, which are available in grocery stores and online.

Overall, the demand for plant-based meat and plant-based milk is increasing in India, as more people become aware of the health and environmental benefits of these products. With more companies entering the market, it is likely that plant-based alternatives will become more widely available and affordable in the future.

1.2 Statement of the problem:

As the demand for meat and milk products continues to rise globally, there is a growing concern about the negative impact of animal husbandry on the environment, animal welfare, and human health. In India, where the majority of the population consumes a vegetarian or lacto-vegetarian diet, there is a growing interest in plant-based alternatives to meat and dairy products.

However, there is little information available about the accessibility, acceptance, nutritional value, sensory qualities, and cost of plant-based meat and milk products in India as compared to conventional animal products. Therefore, there is a need for a comprehensive study to assess the market potential and consumer preferences for plant-based meat and milk products in India, as well as the challenges and opportunities for their production and distribution.

1.3 Need for study

There is a growing need for the study of plant-based milk and plant-based meat in India due to several reasons:

1. **Health concerns:** As people are becoming more health-conscious, they are looking for healthier alternatives to dairy milk and meat products, which are often high in cholesterol, saturated fats, and calories. Plant-based milk and meat products are generally lower in fat, calories, and cholesterol, making them a healthier alternative.
2. **Environmental concerns:** Animal agriculture is a major contributor to greenhouse gas emissions, deforestation, and water pollution. Plant-based milk and meat production have a much lower environmental impact, making them a more sustainable alternative.
3. **Ethical concerns:** Many people are opposed to the treatment of animals in the meat and dairy industry, which often involves animal cruelty and exploitation. Plant-

based milk and meat production do not involve the use of animals, making them a more ethical choice.

4. Economic opportunities: India is the world's largest milk-producing country and has a large vegetarian population. The demand for plant-based milk and meat products is increasing, creating economic opportunities for the development of these industries.

Therefore, it is important to study the production, distribution, and consumption of plant-based milk and meat products in India to better understand the market potential, consumer behaviour, and potential barriers to adoption. This can help in the development of policies and strategies to promote the growth of these industries and provide consumers with more sustainable and ethical alternatives. Keeping in view the above facts, the present study "Analysis of consumer perception and Growth of plant-based meat and milk products" has been designed with the following objectives.

1.4 Objectives

- To elucidate the trends and current market size of plant-based meat and milk products.
- To study the consumer perception towards plant-based meat and milk products.
- To identify the factors influencing the growth of plant-based meat and milk products.

1.5 Scope of the Study

The scope of a study on plant-based meat and plant-based milk in India could include the following aspects:

- Market size and growth potential: The study could explore the current market size of plant-based meat and milk in India, and its growth potential in the coming years.
- Consumer preferences and behaviour: The study could examine the factors that influence consumers to choose plant-based meat and milk over their animal-based counterparts, including taste, health, environmental concerns, and animal welfare.

- **Product availability and variety:** The study could assess the availability and variety of plant-based meat and milk products in India, including their distribution channels.
- **Industry landscape and key players:** The study could analyse the current industry landscape of plant-based meat and milk in India, including the key players.

1.6 Limitations of the study

The sample size is limited, which raises concerns about its generalizability to the broader population and the adoption of convenience sampling may have an impact on the findings by introducing bias.

1. **Statista Intelligence (2021).** "India Plant-based Food Market - Growth, Trends, COVID-19 Impact, and Forecasts (2021 - 2026)". This report provides a comprehensive analysis of the global plant-based protein market, including market size, share, and trends by type, application, and region. It covers key players in the market and examines the growth factors, challenges, and opportunities in the industry.
2. **Global Market Intelligence (2021).** "India Plant-based Food Market - Growth, Trends, COVID-19 Impact, and Forecasts (2021 - 2026)". This report analyzes the growth, trends, and impact of COVID-19 on the plant-based food market in India. It provides market size, share, and forecasts by product type, distribution channel, and region.
3. **Grand View Research, Inc. (2020).** "Plant-Based Food Market Size, Share & Trends Analysis Report". This report offers a comprehensive analysis of the global plant-based food market, including market size, share, and trends by product type, distribution channel, and region.
4. **Cobonre Market Insights (2020).** "India Plant-based Meat Market". This report provides a detailed analysis of the plant-based meat market in India, including market size, share, and trends by product type, source, sales channel, and region. It covers key players in the market and examines the growth factors, challenges, and opportunities in the industry.

CHAPTER 2 - REVIEW OF LITERATURE

Review of literature is important for a study as it helps to identify gaps in current knowledge and research, provides a framework for the research question and hypothesis development and helps to establish the credibility and validity of the study.

3.1 Elucidate the trends and current market size of plant-based meat and milk products

1. **Grand View Research, Inc. (2021).** "Global Plant-Based Protein Market Size, Share & Trends Analysis Report by Type (Soy, Wheat, Pea, Canola), by Application (Food & Beverages, Animal Feed), by Region, and Segment Forecasts, 2021-2028". This report provides a comprehensive analysis of the global plant-based protein market, including market size, share, and trends by type, application, and region. It covers key players in the market and examines the growth factors, challenges, and opportunities in the industry.
2. **Mordor Intelligence (2021).** "India Plant-based Food Market - Growth, Trends, COVID-19 Impact, and Forecasts (2021 - 2026)". This report analyses the growth, trends, and impact of COVID-19 on the plant-based food market in India. It provides market size, share, and forecasts by product type, distribution channel, and region.
3. **Grand View Research, Inc. (2020).** "Plant-Based Food Market Size, Share & Trends Analysis Report". This report offers a comprehensive analysis of the global plant-based food market, including market size, share, and trends by product type, distribution channel, and region.
4. **Coherent Market Insights (2020).** "India Plant-based Meat Market". This report provides a detailed analysis of the plant-based meat market in India, including market size, share, and outlook by product type, source, sales channel, and region. It covers key players in the market and examines the growth factors, challenges, and opportunities in the industry.

3.2 Study the consumer perception towards plant-based meat and milk products

1. **Wilks, M., Phillips, C. J., & Attfield, R. (2020).** Consumer perceptions of plant-based and clean meat in Australia. This Report investigates the attitudes and perceptions of Australian consumers towards plant-based and clean meat products. The study examines factors such as taste, health, and environmental concerns that influence consumer perceptions of plant-based meat products as a viable alternative to traditional meat products.
2. **Poore, J., & Nemecek, T. (2018).** Reducing food's environmental impacts through producers and consumers. This Report discusses the environmental impact of meat production and consumption and the potential role of plant-based meat alternatives in reducing these impacts. The study examines consumer perceptions of plant-based meat products as a more sustainable and environmentally friendly alternative to traditional meat products.
3. **Sogari, G., Mora, C., Menozzi, D., & Mannerini, T. (2017).** Exploring the determinants of attitudes and behaviours towards meat alternatives among Italian consumers. This Report investigates the attitudes and behaviours of Italian consumers towards meat alternatives, including plant-based meat products. The study examines factors such as health concerns, ethical considerations, and taste preferences that influence consumer perceptions of plant-based meat products.
4. **de Boer, J., Schösler, H., & Aiking, H. (2014).** Meatless days and less meat consumption: Exploring socio-psychological determinants of meat consumption reduction. This Report explores the psychological and social factors that influence meat consumption reduction, including the adoption of plant-based meat alternatives. The study examines factors such as attitudes, habits, social norms, and values that affect consumer perceptions of plant-based meat products.

3.3 Identify the factors influencing the growth of plant-based meat and milk products

1. **Haddad, C., & Elten, M. (2020).** The growing market for plant-based meat and dairy alternatives: drivers, developments, and challenges. *International Journal on Food System Dynamics*. This Study examines the drivers behind the growing market for plant-based meat and dairy alternatives, including health concerns, environmental sustainability, and ethical considerations. The review also explores the challenges facing the industry, such as consumer acceptance, cost, and taste.
2. **De Backer, C. J., Hudders, L., & Pauwels, L. (2020).** This Study investigates the drivers and barriers of plant-based meat consumption in Belgium. The review identifies several factors that influence consumer behaviour, including taste, health concerns, environmental sustainability, and social norms.
3. **Smetana, S., & Mathys, A. (2020).** Plant-based meat substitutes in the flexitarian age: the right time for pea protein. This Report examines the factors that have led to the growth of plant-based meat substitutes, including increasing consumer awareness of the environmental and health impacts of meat consumption. The study also highlights the potential of pea protein as a sustainable and nutritious alternative to animal-based protein sources.

15 **Research Design:** Cross-sectional survey design was used to collect data from participants in the four cities. This design allowed us to collect data at a single point in time and compare responses across four cities.

16 **Sampling technique:** Convenience sampling technique was used to select participants. Convenience sampling involves selecting participants who are readily available and willing to participate in the study.

CHAPTER 3 -RESEARCH METHODOLOGY

This chapter deals with the selection of study area, location, sampling procedure, variables, research design, method of data collection and statistical tools used for analysis. These details are presented in the following analysis.

- 2.1 Selection of study area: The data is collected from four major cities in India - Bangalore, Mumbai, Chennai, and Delhi.
- 2.2 Location of study area: The four cities that is selected for study are geographically diverse and represent major urban centres in different regions of India. Bangalore is located in the southern part of the country, Mumbai in the western part, Chennai in the south-eastern part, and Delhi in the northern part of India. These cities are known for their cosmopolitan and diverse populations and are likely to have a higher demand for plant-based products.
- 2.3 Sampling procedure: The target population was Tier-1 city residents due to diversity and high demand for plant-based products. Convenience sampling was used to collect 100 responses that allows to access participants quickly and easily.
- 2.4 Variables and their measurement: The demographic variables used in the study are following Age, location, occupation, Income, Education. Age and Income is a continuous variable, whereas location, occupation and education are categorical variables.
- 2.5 Research Design: Cross-sectional survey design was used to collect data from participants in the four cities. This design allowed us to collect data at a single point in time and compare responses across four cities.
- 2.6 Sampling technique: Convenience sampling technique was used to select participants. Convenience sampling involves selecting participants who are readily available and willing to participate in the study.

2.7 Data Collection method: Questionnaire was used to collect data from participants etc. Secondary data was collected through various literature, online databases, company reports, etc.

2.8 Data Analysis and statistical tools applied: Descriptive statistics were used to analyse data. Descriptive statistics involves summarizing and describing data using measures such as:

Percentage: Percentage is a way of expressing a number as a fraction of 100. It is often denoted by the symbol "%".

The formula for calculating a percentage is:

$$\text{Percentage} = (\text{Part} / \text{Whole}) \times 100$$

Median: The median is a measure of central tendency in statistics, which represents the middle value in a set of data when arranged in order from smallest to largest.

To find the median of a set of data:

- 1) Arrange the data in order from smallest to largest.
- 2) If the number of data points is odd, the median is the middle value.
- 3) If the number of data points is even, the median is the average of the two middle values.

Frequency: Frequency is a measure of how often a particular value or observation occurs in each set of data. It is commonly used in statistical analysis and research.

The formula for calculating frequency is:

Frequency = Number of times a particular value or observation occurs in a data set

Chi-square was also used to see the association between categorical variables.

Chi-square test- A chi-square test is a statistical test used to determine whether there is a significant association between two categorical variables. The formula for the chi-square test depends on the specific type of chi-square test being used. However, the general formula for calculating the test statistic is:

$$\chi^2 = \sum (O - E)^2 / E$$

Where:

- χ^2 is the test statistic
- O is the observed frequency in each category of one variable
- E is the expected frequency in each category, assuming no association between the two variables
- $\sum$ is the sum of all categories

The expected frequency, E, for each category is calculated using the formula:

$$E = (\text{row total} * \text{column total}) / \text{total}$$

Where:

- row total is the total number of observations in a row
- column total is the total number of observations in a column
- total is the total number of observations in the entire table

The chi-square test statistic measures the difference between the observed and expected frequencies, and it is calculated by summing up the squared differences between the observed and expected frequencies, divided by the expected frequencies. The resulting test statistic follows a chi-square distribution with degrees of freedom equal to $(R-1) * (C-1)$, where R is the number of rows and C is the number of columns in the contingency table.

CHAPTER 4 - RESULTS AND DISCUSSION

4.1 Elucidate the trends and current market size of plant-based meat and milk products

4.1.1 Global Market Overview: The worldwide marketplace for plant-primarily based total meat, which become predicted to be worth \$6.62 billion in 2020, is anticipated to increase to \$33.99 billion by 2027, with a CAGR of 15%. The biggest markets for meat substitutes are the United States, the United Kingdom, and China. Beyond Meat, Impossible Foods, and Quorn are significant market participants.

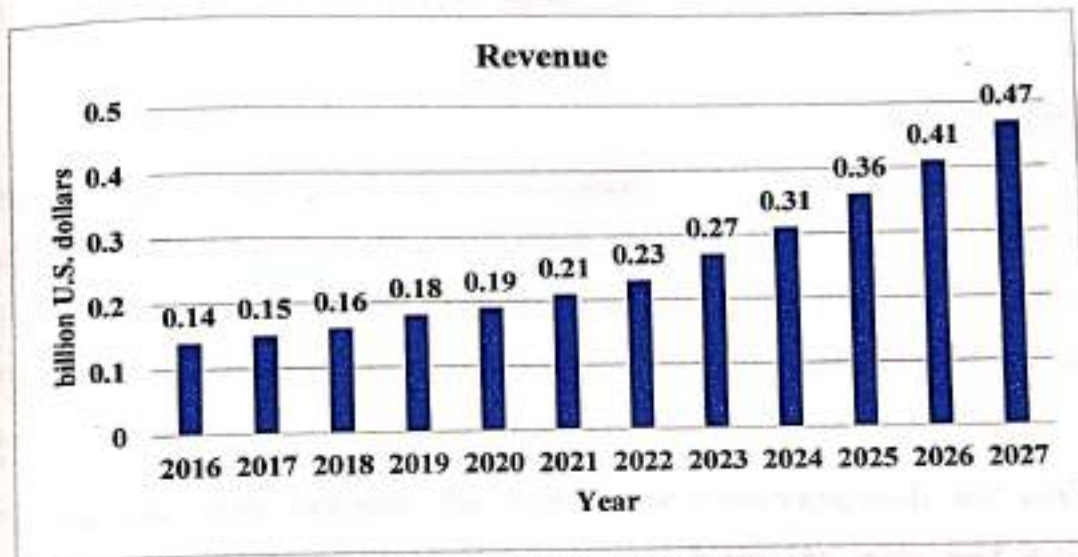


Figure 1: Revenue of plant-based meat globally

Source- Statista, 2023

In 2020, the market for plant-based milk was estimated to be worth \$17.97 billion, and by 2027, it is expected to reach \$44.2 billion. The greatest market for milk alternatives is in the Asia-Pacific area, followed by Europe and North America. Soy milk, almond milk, and coconut milk are the most often used milk alternatives. Oat milk, hemp milk, and rice milk are some more milk alternatives. Those who are lactose intolerant, vegan, or have other dietary limitations frequently use these replacements.

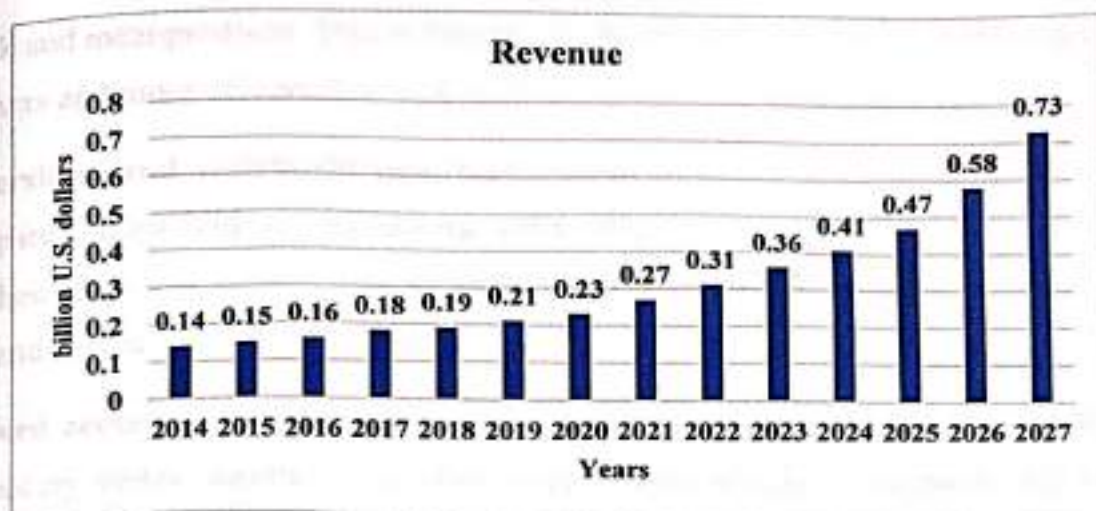


Figure 2: Revenue of plant-based milk globally

Source- Statista, 2023

4.1.2 Trends in the global plant-based food market

According to a survey by Meticulous Research, there has been a discernible shift towards plant-based substitutes for conventional dairy and meat products in recent years. Following are some of the major developments in the global markets for plant-based milk and meat:

1. Increasing consumer demand: The demand for plant-based milk and meat products has significantly increased. Plant-based foods are becoming more and more popular among consumers for ethical, environmental, and health grounds.

2. Product portfolio diversification: Producers of plant-based milk and meat are adding more goods to their product portfolios. New flavours, formats, and ingredients that satisfy various consumer preferences are included in this.

3. Increased availability: Plant-based milk and meat products are becoming more widely available in supermarkets, restaurants, and other food outlets. This is making it easier for consumers to access and try these products.

4. Investment in research and development: Companies are investing heavily in research and development to improve the taste, texture, and nutritional value of plant-

based milk and meat products. This is helping to create products that are more appealing to consumers and more competitive with traditional dairy and meat products.

5. Partnerships and collaborations: Plant-based milk and meat manufacturers are forming partnerships and collaborations with other companies and organizations to increase their reach and impact. This includes collaborations with food service providers, retailers, and agricultural organizations.

6. Improved accessibility: Vegetarian milk and meat alternatives are getting easier to find in grocery stores, eateries, and other food establishments. Customers will find it simpler to obtain and sample these products as a result.

7. Investment in R&D: Businesses are making significant R&D investments to enhance the flavour, texture, and nutritional content of goods made from plant milk and meat. As a result, products are becoming more consumer-friendly and competitive with conventional dairy and meat goods.

8. Collaborations and alliances: To broaden their influence and market, producers of plant-based milk and meat are forging alliances and alliances with other businesses and organizations. Collaborations with retailers, food service providers, and agricultural groups fall under this category.

Overall, the global plant-based milk and meat markets are expected to continue to grow in the coming years as consumers become increasingly interested in plant-based options and manufacturers continue to innovate and improve their products.

As per Statista 2023, The average revenue generated by plant-based meat products per person has not increased significantly over the past 13 years. From the year 2014 to the year 2019, the growth has remained flat. It increased in 2020 and then again in 2022. The average revenue growth is expected to remain steady between 2022 and 2026.

4.1.3 Plant-based meat

Indian Scenario: The market for plant-based meat products in India was valued at approximately \$25 million in 2020. The market is expected to grow at a CAGR of around 10% from 2021 to 2026. India is one of the largest consumers of meat globally, with a significant vegetarian and vegan population. The increasing awareness about the benefits of a plant-based diet, coupled with concerns about animal welfare and the environment, is driving the growth of the plant-based meat market in the country.

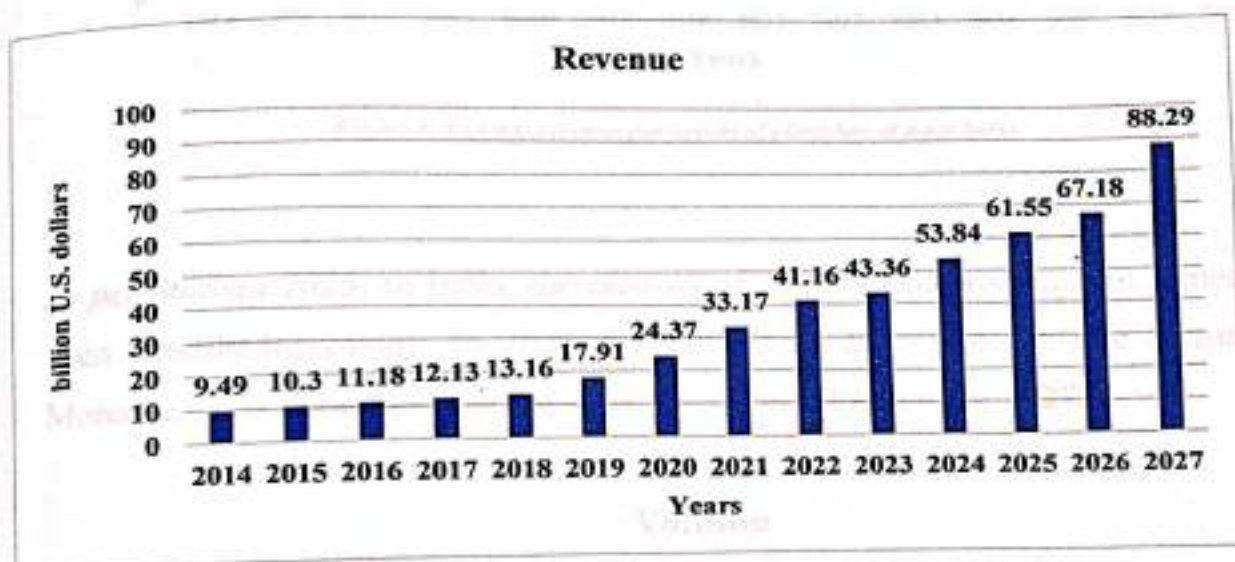


Figure 3: Revenue of plant-based meat India

Source- Statista, 2023

As per Statista 2023, The **average revenue** generated by plant-based meat products per person has not increased significantly over the past 13 years. From the year 2014 to the year 2019, the growth has remained flat. It increased in 2020 and then again in 2022. The average revenue growth is expected to remain steady between 2022 and 2026.

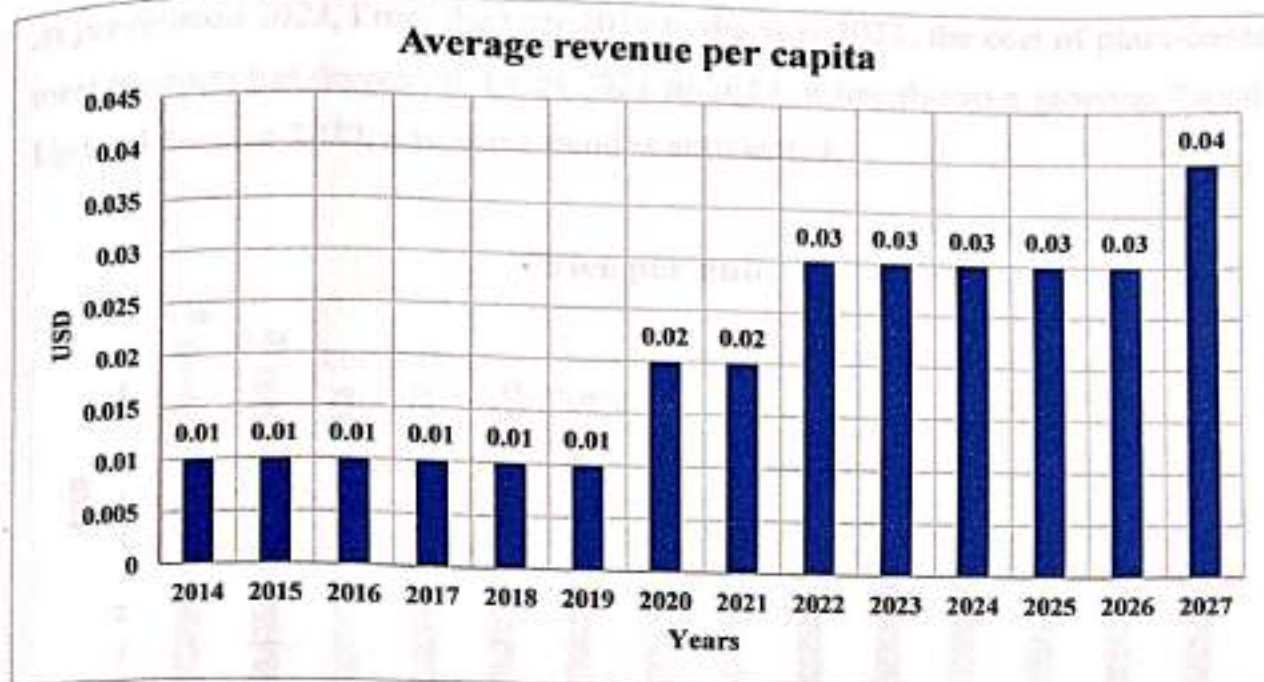


Figure 4: Average revenue per capita of plant-based meat India

As per Statista 2023, In India, the amount of vegan meat products consumed has been steadily increasing. In 2023, 6.7 million kg of volume will be consumed. Moreover, it is anticipated to reach 11.6 million kilograms in 2027.

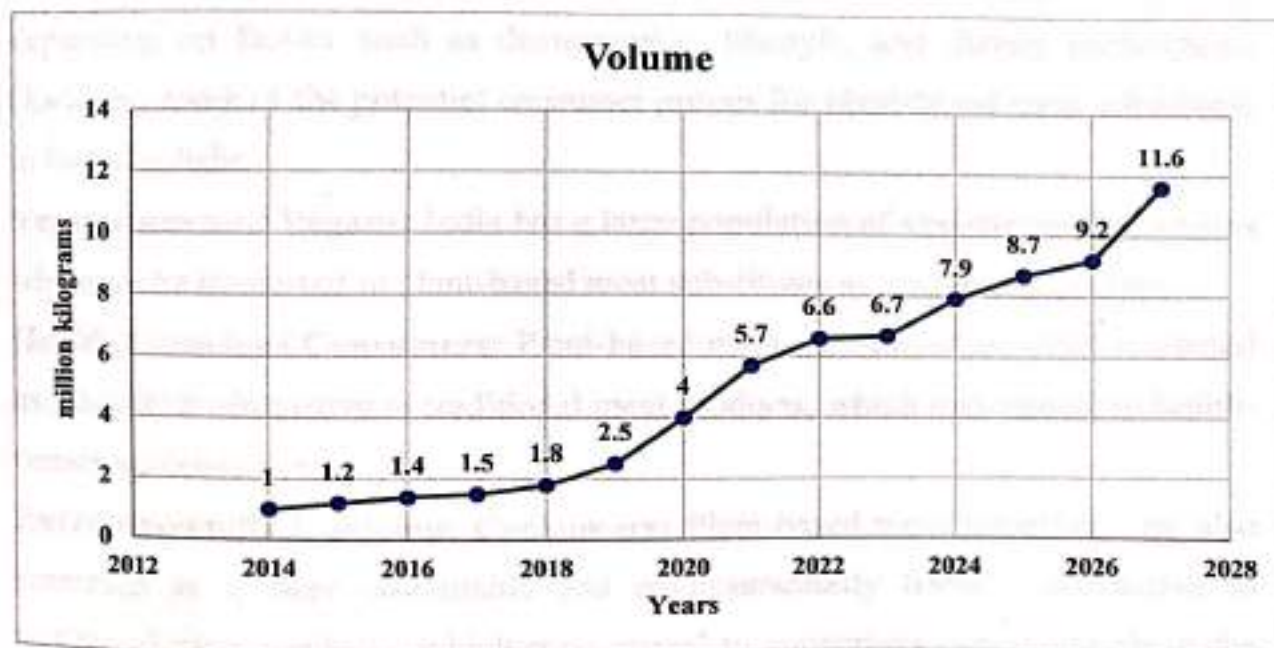


Figure 5: Volume of plant-based meat India

As per Statista 2023, From the year 2014 to the year 2021, the cost of plant-based meat products has decreased. From 2021 to 2023, it has shown a growing Trend. Up until the year 2027, a positive trend is anticipated.



Figure 6: Price per unit of plant-based meat India

Target Consumer Groups for Plant-Based Meat Substitutes-

The target consumer groups for plant-based meat substitutes in India may vary depending on factors such as demographic, lifestyle, and dietary preferences. However, some of the potential consumer groups for plant-based meat substitutes in India include:

1. **Vegetarians and Vegans:** India has a large population of vegetarians and vegans who may be interested in plant-based meat substitutes as a source of protein.
2. **Health-Conscious Consumers:** Plant-based meat substitutes are often marketed as a healthier alternative to traditional meat products, which may appeal to health-conscious consumers.
3. **Environmentally Conscious Consumers:** Plant-based meat substitutes are also promoted as a more sustainable and environmentally friendly alternative to traditional meat products, which may appeal to consumers concerned about the impact of animal agriculture on the environment.

4. **Flexitarians:** Flexitarians are consumers who primarily follow a plant-based diet but occasionally consume meat. Plant-based meat substitutes may be a convenient and tasty option for these consumers to incorporate into their diets.
5. **Younger Consumers:** Younger consumers, including millennials and Gen Z, are increasingly interested in plant-based products and maybe a key target market for plant-based meat substitutes.

Market segmentation by product- Based on Imarc group *Report*, the plant-based meat market is segmented based on Source segmentation, Type segmentation in the plant-based Meat Market, Segmentation by types of products, Distribution channels, and storage.

- **Source segmentation** – Soy, wheat, Blends, Peas, and other sources (quinoa, oats, beans, nuts, and seeds)
- **Type segmentation** – Beef, Chicken, Pork, Fish
- **Types of products-** Burger patties, Strips & nuggets, Sausages, Meatballs, Other products (slices, fillets, cutlets, slides, figures, and crumbles)
- **Distribution channels-** Food retail, food services, E-commerce
- **By storage-** Frozen, Refrigerated, shelf-stable

Key market players-

The plant-based meat industry in India is still in its nascent stages, and there are only a few players operating in this space. Some key market players in the plant-based meat industry in India include:

1. **Good Dot:** Good Dot is a plant-based meat company that offers a range of products such as vegetarian chicken, vegetarian mutton, and vegetarian keema.
2. **Veggie Champ:** Veggie Champ is a plant-based meat brand that offers a range of products such as vegetarian chicken, vegetarian mutton, and vegetarian fish.
3. **Blue Tribe Foods:** Blue Tribe Foods is a plant-based meat company that offers a range of products such as vegetarian chicken nuggets, vegetarian chicken popcorn, and vegetarian kebabs.
4. **Nutriplato:** Nutriplato is a plant-based food company that offers a range of products such as vegetarian meatballs, vegetarian burgers, and vegetarian sausages.
5. **Vezlay Foods:** Vezlay Foods is a plant-based food company that offers a range of products such as vegetarian chicken, vegetarian mutton, and vegetarian seekh kebab.

4.1.4 Plant-based dairy products: The market for plant-based dairy was valued at 0.23 billion U.S. dollars in 2020 and is expected to grow at a CAGR of 22.4% from 2021 to 2027. It also states that the demand for plant-based dairy products is expected to increase due to factors such as the rising number of lactose-intolerant consumers, increasing health consciousness, and growing adoption of veganism and vegetarianism.

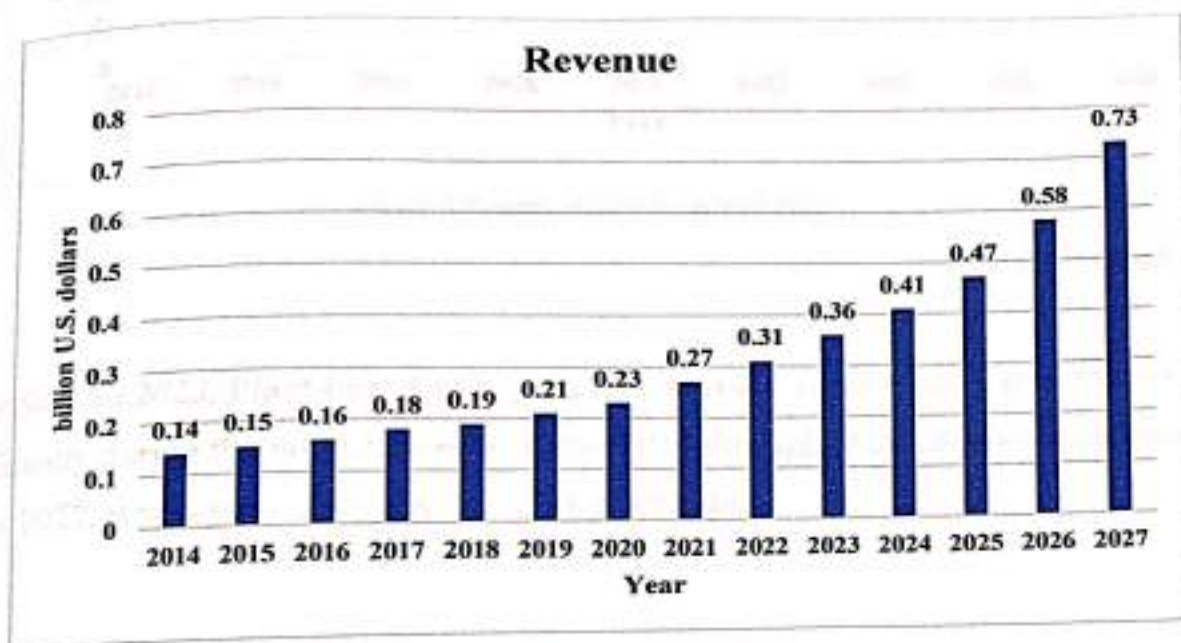


Figure 7: Revenue of plant-based milk India

Source- Statista, 2023

As per Statista 2023, In India, the amount of plant-based milk products consumed has increased steadily. In 2023, 225.6 million kilograms of volume will be consumed, and in 2027, 391.2 million kilograms are anticipated.

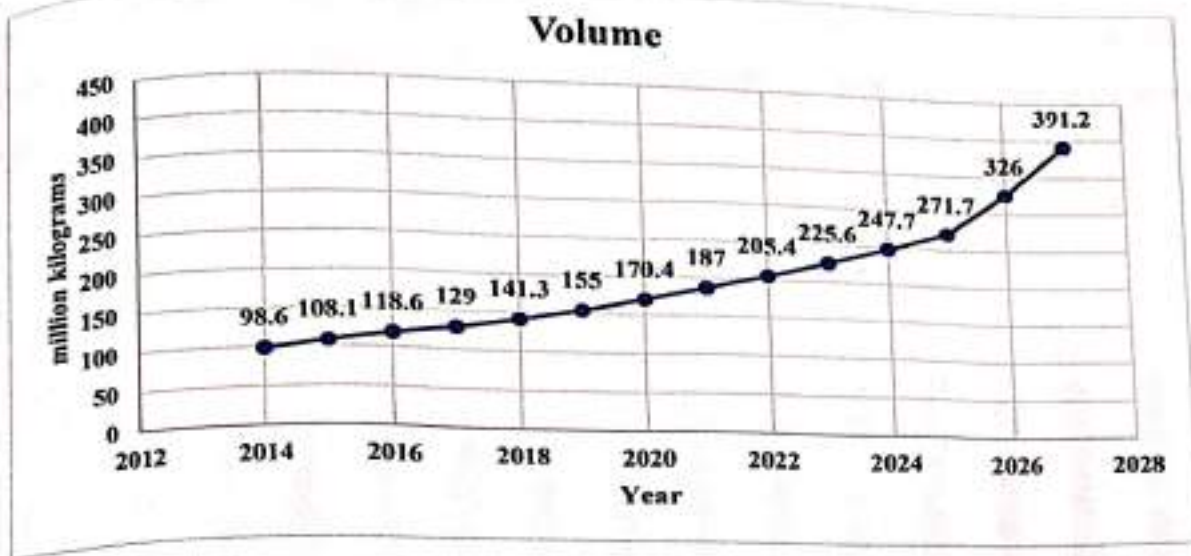


Figure 8: Revenue of plant-based milk India

Source- Statista, 2023

As per Statista 2023, Plant-based milk products' average volume per capita has increased significantly during the last 12 months. From 2014 through 2022, the rise was consistent, and by 2027, it was predicted to reach 0.3 kg per capita.

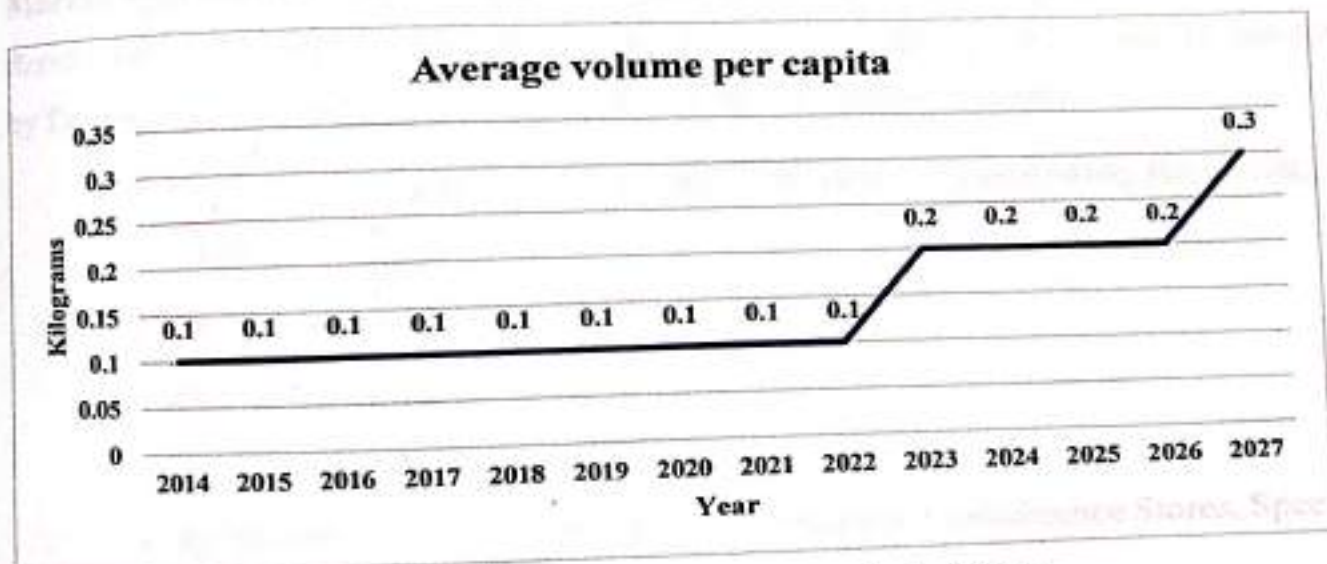


Figure 9: Average volume per capita of plant-based milk India

Source- Statista, 2023

As per Statista 2023, Throughout the years 2014 and 2020, there was a downward tendency in the price of milk products made from plants. From 2021 onward, a positive trend is anticipated



Figure 10: Price per unit of plant-based milk India

Market segmentation by product- As per Meticulous Research Report 2022 & FMI Report 2022, Plant-based milk products can be segmented by type of product, by category, by flavour, by formulation of the product and by distribution channel.

- **By Type-** Coconut Milk, Soy Milk, Oat Milk, Almond Milk, Rice Milk, and Others.
- **By Category-** Organic, Conventional
- **By Formulation -** Liquid, Powder
- **By flavour-** Unflavoured, Flavoured
- **By Distribution Channel-** Modern Groceries, Convenience Stores, Specialty Stores, Online Retail, Others

Target Consumer Groups for Plant-Based Milk Substitutes-

Plant-based milk substitutes in India target various consumer groups, including:

1. **Vegans and Vegetarians:** Plant-based milk substitutes are an ideal option for vegans and vegetarians who avoid animal-derived products for ethical, environmental, or health reasons.
2. **Lactose-Intolerant Consumers:** Lactose intolerance is prevalent in India, and plant-based milk substitutes offer a lactose-free alternative to traditional dairy products.
3. **Health-Conscious Consumers:** Plant-based milk substitutes are low in saturated fat and cholesterol, making them a healthier alternative to traditional dairy milk. They are also rich in nutrients and minerals.
4. **Consumers with Allergies:** Some consumers may have allergies to dairy or soy products, and plant-based milk substitutes offer a safe and healthy alternative for them.
5. **Environmentally Conscious Consumers:** Plant-based milk substitutes have a lower environmental impact than traditional dairy products, making them a popular choice among environmentally conscious consumers.
6. **Consumers Looking for Variety:** Plant-based milk substitutes offer a variety of flavours, such as almond milk, coconut milk, and oat milk, which appeal to consumers looking for variety in their diets.

Overall, the market for plant-based milk substitutes in India is growing rapidly, driven by various consumer groups who are looking for healthier, more sustainable, and ethical alternatives to traditional dairy products.

Key market players-

Some key market players in the plant-based milk industry in India include:

1. **Goodmylk:** Goodmylk is a plant-based milk company that offers a range of milk alternatives such as almond milk, coconut milk, and oat milk.
2. **Raw Pressery:** Raw Pressery is a health and wellness brand that offers a range of plant-based beverages, including almond milk and coconut milk.
3. **Urban Platter:** Urban Platter is an online store that offers a variety of plant-based milk alternatives such as almond milk, soy milk, and oat milk.
4. **Sofit:** Sofit is a popular brand that offers soy milk in different flavors like vanilla, chocolate, and mango.
5. **NutriSoy:** NutriSoy is a brand that offers a range of soy milk products such as plain soy milk, vanilla soy milk, and chocolate soy milk.
6. **Plant Power:** Plant Power is a brand that offers almond milk, oat milk, and soy milk in different flavors like vanilla and chocolate.
7. **Tofuji:** Tofuji is a brand that offers a range of soy products, including soy milk, tofu, and soy paneer.
8. **The Butternut Co.:** The Butternut Co. is a brand that offers almond milk and cashew milk in different flavors like unsweetened and vanilla.

4.2 Consumer perception towards plant-based meat and milk products

A total of 101 Responses were collected through a Google form survey. The data was analysed using MS Excel & Tableau. Attributes like Demographic profile, awareness about Plant-based Products, Preference of product, Preference of Brand, Factors influencing the Purchase of Plant-Based products, and Distribution channel.

4.2.1 Location of Respondents

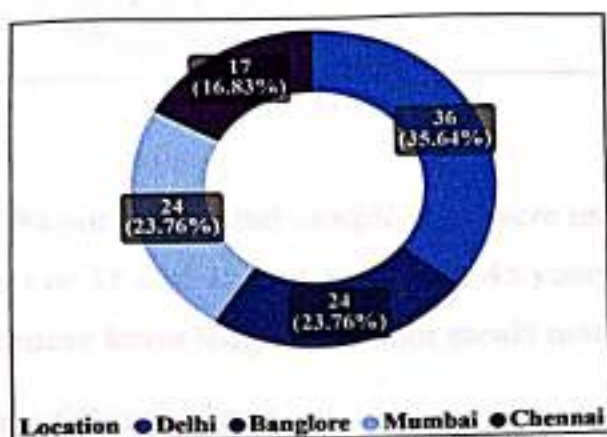


Figure 11: Location

From figure 11, it can be interpreted that Delhi had 36% of the respondents, followed by Bangalore (23%), Mumbai (23%), and Chennai (16%).

4.2.2 Occupation of Respondents

The figure 12 revealed that 64% of the total sample size were students followed by self-employed people (22%). Government employees and others also responded to the questions.

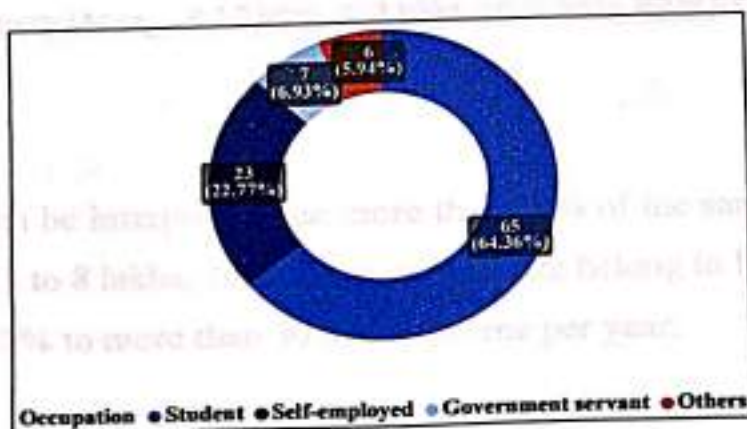


Figure 12: Occupation

4.2.3 Age of Respondents

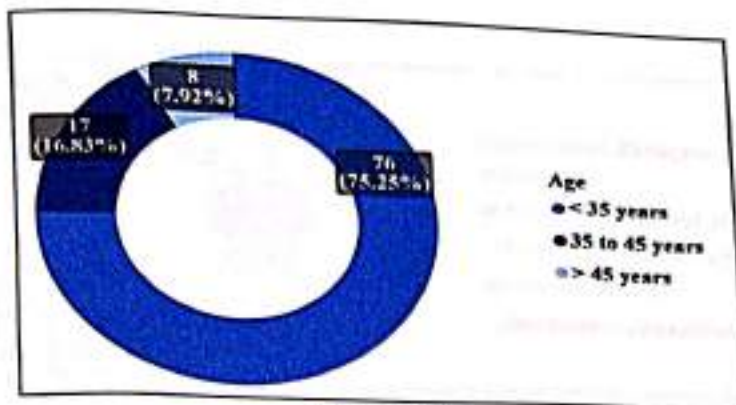


Figure 13: Age

Figure 13 depicts that 75% out of the total sample size, were under the age of 35, followed by 17% between the ages of 35 and 45 and 8% above 45 years. According to the survey, younger individuals are more knowledgeable about meals made from plants.

4.2.4 The annual income of Respondents

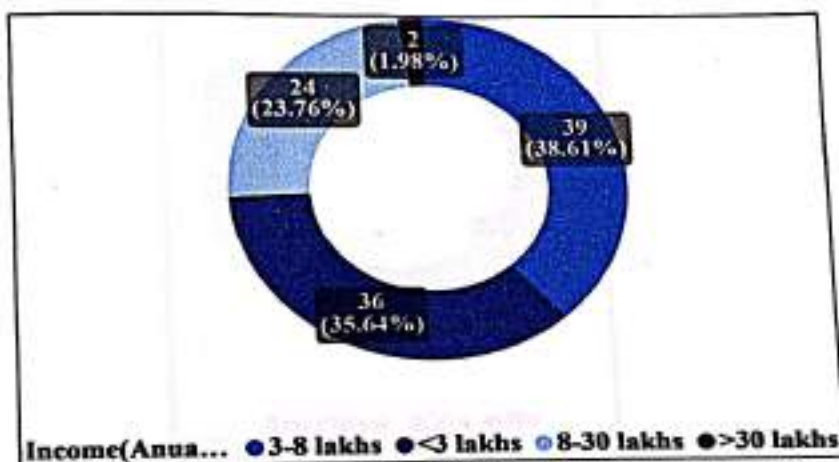


Figure 14: Annual Income

From figure 14, it can be interpreted that more than 39% of the sample size belonged to the income range of 3 to 8 lakhs, 36% of the sample size belong to less than 3 lakhs, 23% to 8 to 30 lakhs and 2% to more than 30 lakhs income per year.

4.2.5 Educational background of Respondents

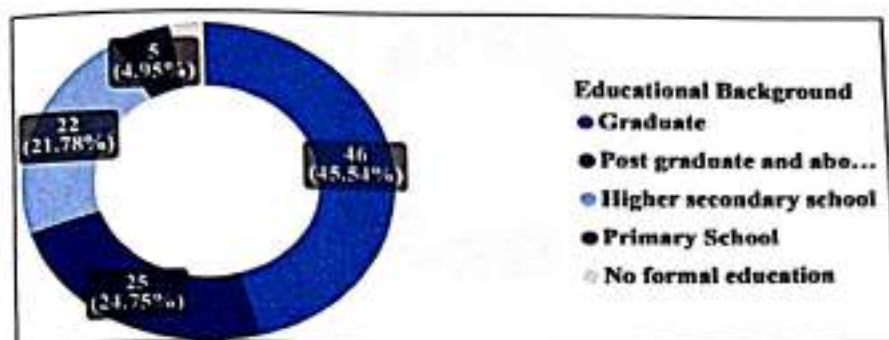


Figure 15: Educational Background

From figure 15, it can be interpreted that 45% of respondents are having a graduate degree, 24% of them are having post graduate degree and above. Respondents were also from higher secondary school (22%), Primary school (5%) and no formal education (4%).

4.2.6 Awareness of Respondents



Figure 16: Awareness

From figure 16, it can be interpreted 68% of the respondents were aware about plant-based meat and milk products leaving the rest.

4.2.7 Monthly Expenditure

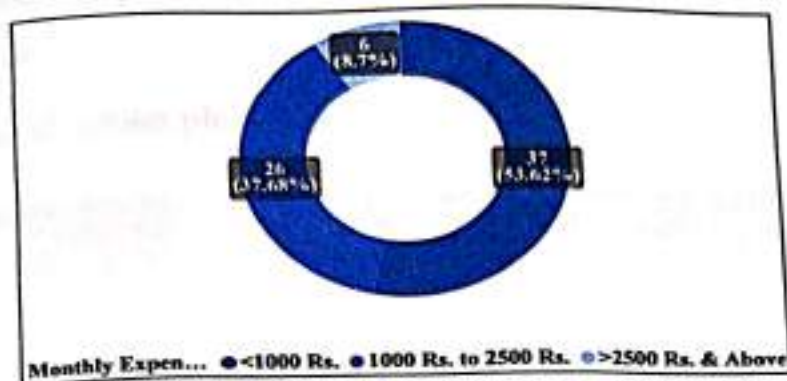


Figure 17: Monthly Expenditure

From figure 17, it can be interpreted that more than 50% of respondents spent less than 1000 rupees on plant-based food products, 37% spends 1000 to 2500 rupees and 8% spends more than 2500 rupees.

4.2.8 Distribution channel of plant-based milk and meat products

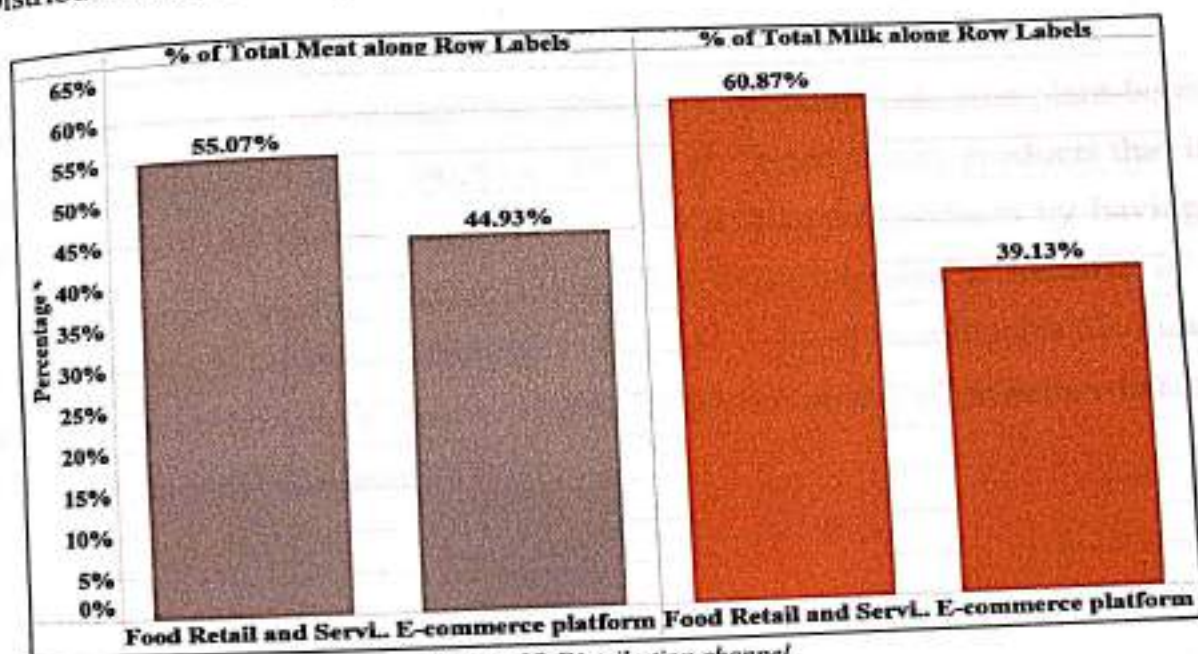


Figure 18: Distribution channel

From figure 18, it can be interpreted that 55% of respondents use food retail and services as distribution channel for plant-based meat, 60% of them use Food retail & services for plant-based milk also.

4.2.9 Reason behind choosing plant-based products



Figure 19: Reason behind choosing plant-based products

From figure 19, it can be interpreted that 29% of respondents consume plant-based food because of veganism, as vegans are those who don't consume any products that involve animals, they can have similar experiences of animal-based products by having plant-based products. 25% of respondents choose plant-based products because of animal welfare as butchery is a direct suffering to animals. 25% of respondents also see health consciousness as a reason, as vegans do suffer from a deficiency of protein, whereas plant-based food can be a great source of it.

4.2.10 Chi-square test

The chi-square test has been conducted to see if there is any significant association of Age, Location, Education, Occupation, and Income with the Awareness of respondents of plant-based food.

Table 1: Chi-square test

Variable	Level of significance	χ^2	p-value
Age	0.05	8.049	0.017
Location	0.05	2.409	0.491
Education	0.05	5.761	0.217
Occupation	0.05	1.346	0.718
Income	0.05	8.049	0.944

In the **First case**, the variables are **age and awareness**.

The chi-square value measures the difference between the observed frequencies and the expected frequencies under the null hypothesis that there is no association between the two variables. The p-value is the probability of observing a chi-square value as extreme or more extreme than the one observed, assuming the null hypothesis is true.

Since the p-value is less than the significance level of 0.05, we can reject the null hypothesis and conclude that **there is a significant association between age and awareness**. In other words, the two variables are not independent and there is evidence of a relationship between them.

In **Second case**, the variables are **location and awareness**.

The null hypothesis for the chi-square test is that there is no significant association between the two variables, and the alternative hypothesis is that there is a significant association. The p-value of 0.491 is greater than the level of significance of 0.05, which suggests that there is **not a statistically significant association between the location and awareness** variables. In other words, we fail to reject the null hypothesis and

conclude that there is no evidence to suggest that there is a relationship between the location and awareness variables. It is important to note that a non-significant result does not necessarily mean that there is no relationship between the two variables. It simply means that we cannot reject the null hypothesis based on the available data. Other factors, such as sample size and statistical power, may also have an impact on the results of the chi-square test.

In **Third case**, the variables are **Education and awareness**

The p-value of 0.217 is greater than the level of significance of 0.05, which suggests that there is **not a statistically significant association between education and awareness**. In other words, we fail to reject the null hypothesis and conclude that there is no evidence to suggest that there is a relationship between the education and awareness variables. However, it is important to note that a non-significant result does not necessarily mean that there is no relationship between the two variables. It simply means that we cannot reject the null hypothesis based on the available data. Other factors, such as sample size and statistical power, may also have an impact on the results of the chi-square test.

In **Fourth case**, the variables are **Occupation and awareness**

The p-value of 0.718 is greater than the level of significance of 0.05, which suggests that there is **not a statistically significant association between occupation and awareness**. In other words, we fail to reject the null hypothesis and conclude that there is no evidence to suggest that there is a relationship between the occupation and awareness variables.

In **Fifth case**, the variables are **Income and awareness**

The null hypothesis for this test is that there is no significant association between the two variables. The p-value of 0.944 indicates the probability of observing a test statistic as extreme as the one calculated from the data, assuming that the null hypothesis is true. In this case, the p-value is much larger than the level of significance of 0.05. Since the p-value is greater than the level of significance, we fail to reject the null hypothesis. Therefore, we can conclude that there is not enough evidence to suggest that there is a

significant association between income and awareness. In other words, we cannot say that awareness levels differ significantly across different income groups based on the data. From Chi-square test we can suggest that the distribution of awareness levels across different age groups is not due to chance. In other words, there is a relationship between these two variables, and age may be a significant predictor of awareness.

4.2.11 Preference for plant-based meat products

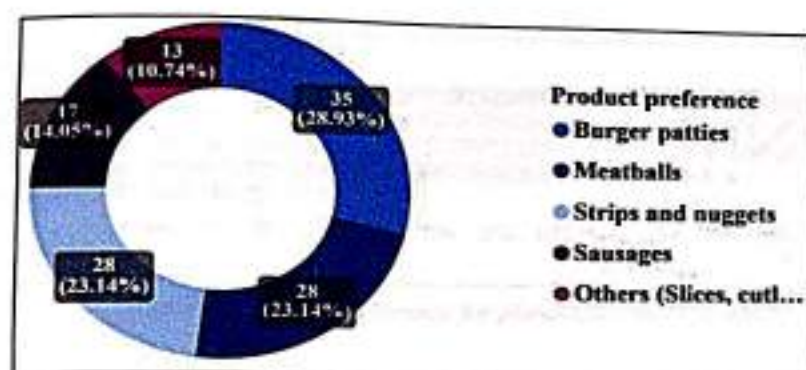


Figure 20: Preference for plant-based meat products

From figure 20, it can be interpreted that most preferred product of plant-based meat in the sample size surveyed is Burger patties, 23% of respondents prefer meatballs, followed by strips & nuggets and sausages. 10% of respondents also prefer slices, cutlets etc.

4.2.12 Preference for plant-based meat brands

Table 2: Preference for plant-based meat brands

Imagine Meats	28.10%
Wakao	22.31%
Evo Foods	22.31%
GoodDot	16.53%
Vezlay	10.74%

From Table 2, it can be interpreted that most of the respondents prefer the newly entered startups like Imagine Meats (based out of Mumbai), followed by Wakao (based out of Mumbai) as these startups give a wide range of services, a wide range of products and do sufficient promotional activities to attract the young generation. 26% of respondents do prefer big brands like GoodDot & Vezlay.

4.2.12 Source preference for plant-based meat products

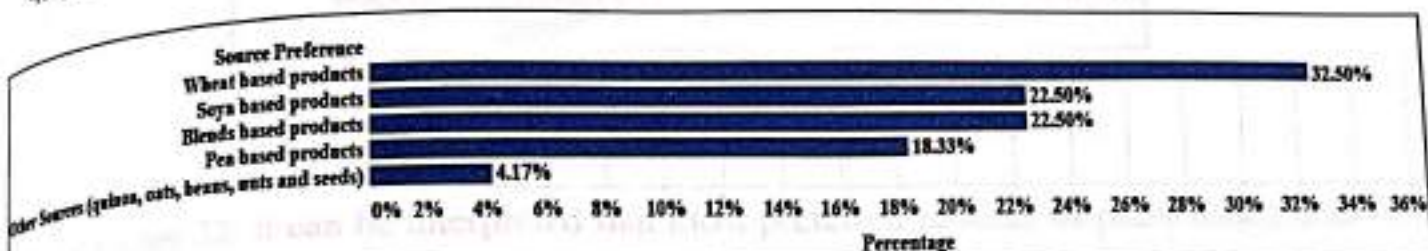


Figure 21: Source preference for plant-based meat products

From Figure 21, it can be interpreted that 32% of the respondents prefer wheat-based products, followed by 22% of them prefer soya-based products which are mostly consumed in the form of plant-based meat. Some other sources like Blends, pea-based are also preferred by large audiences. Respondents do prefer products made from quinoa, oats, beans, nuts, and seeds as well.

4.2.13 Preference for plant-based milk products

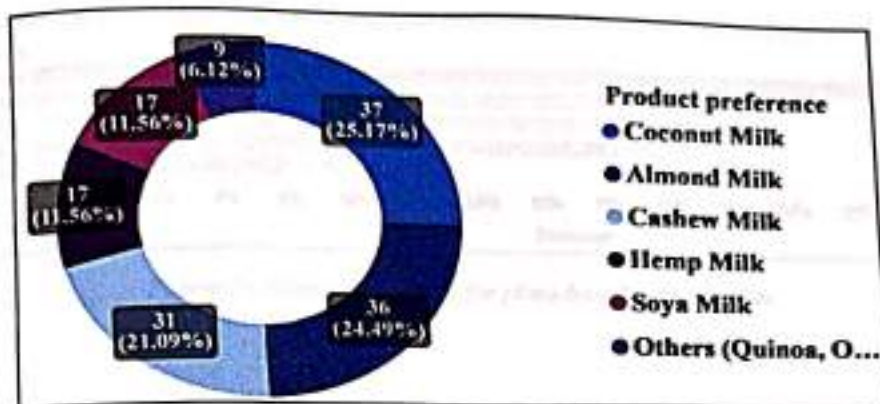


Figure 22: Preference for plant-based milk products

From Figure 22, it can be interpreted that most preferred product of plant-based milk in the sample size surveyed is Coconut milk, 24% of respondents prefer Almond milk, followed by cashew milk and Hemp milk. 17% of respondents also prefer Soya milk.

4.2.14 Preference for plant-based milk brands

Table 3: Preference for plant-based milk brands

Urban Platter	26.92%
Nutriva	26.92%
Raw Pressery	18.46%
Sofit	14.62%
Goodmylk	11.54%
Others	1.54%

From Table 3, it can be interpreted most of the respondents prefer the newly entered startups like Urban Platter (based out of Mumbai), followed by Nutriva (based out of Mumbai) as these startups give a wide range of services, a wide range of products and do sufficient promotional activities to attract the young generation. 18% of respondents prefer raw Pressery brand and 14% of them prefer Sofit which is Hershey India's Brand.

4.2.15 Source preference for plant-based milk products

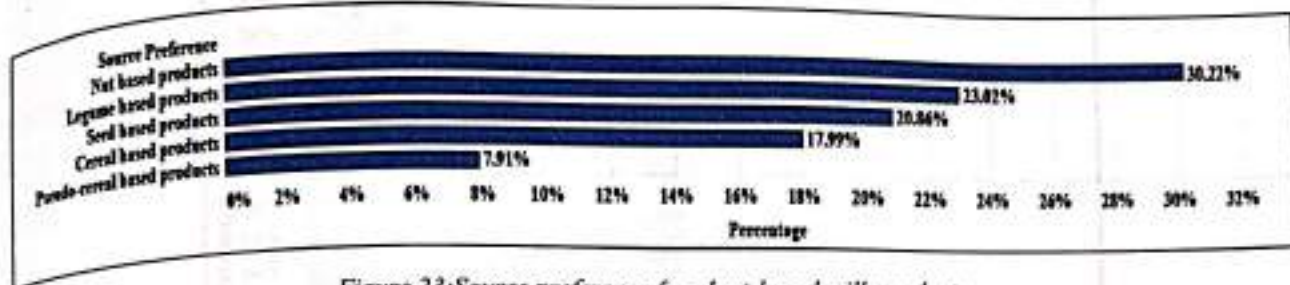


Figure 23: Source preference for plant-based milk products

From Figure 23, it can be interpreted that 30% of the respondents prefer nut-based products like almonds & cashew, followed by 23% of them prefer legume-based products. Some other sources like seed-based, and cereal-based like Rice are also preferred by large audiences. Respondents do prefer products made from pseudo-cereal like quinoa as well.

4.3 factors influencing the growth of plant-based meat and milk products

4.3.1 Factors influencing the growth of plant-based meat

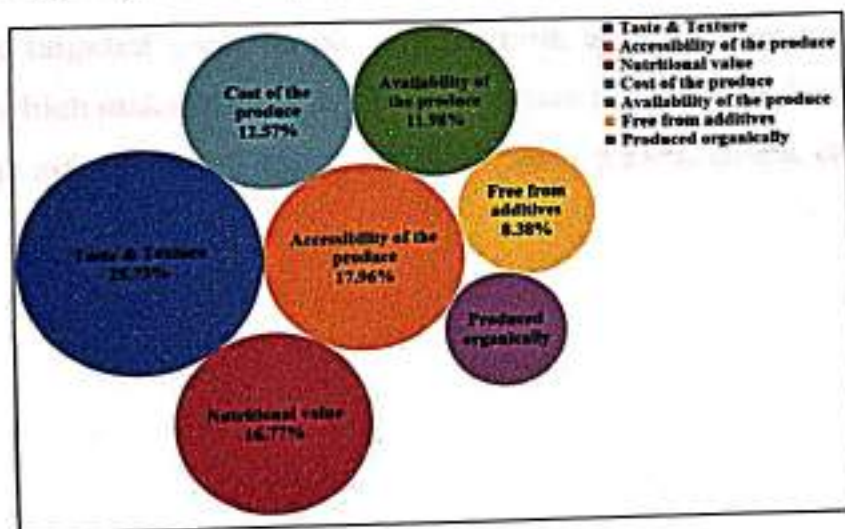


Figure 24: Factors influencing the growth of plant-based meat

From Figure 24, it can be interpreted that 25% of the respondents prefer Taste & Texture of the food product followed by Accessibility of produce (17%), Nutritional value (16%), cost of the produce (12%), availability of the produce (11%) and free from additives (8%).

4.3.2 Factors influencing the growth of plant-based milk

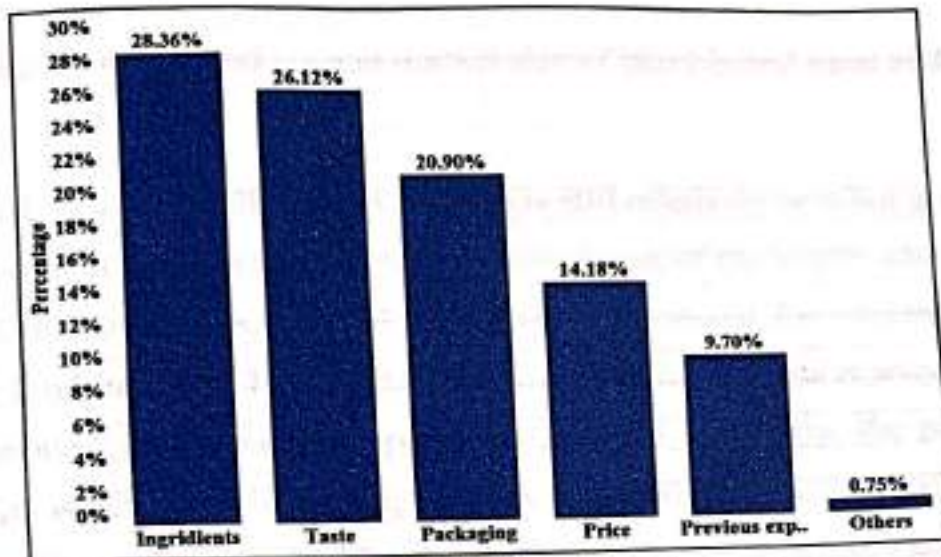


Figure 25: Factors influencing the growth of plant-based meat

From Figure 25, it can be interpreted that 28% of the respondents prefer Ingredients of the milk products followed by Taste (26%), packaging (20%), price (14%), and previous experience (9%). The targeted group in plant-based milk is the lactose intolerance and sustainability group, which makes the ingredients important for purchase for this segment, as they don't want any animal-based milk traces, additives, preservatives, etc.

CHAPTER 5 – CONCLUSION AND RECOMMENDATIONS

5.1 To elucidate the trends and current market size of plant-based meat and milk products

The plant-based meat and milk market in India is still relatively new but growing rapidly. In recent years, there has been an increasing awareness of the health and environmental benefits of plant-based diets, as well as a growing demand for vegetarian and vegan options in the food industry. The plant-based meat market in India is projected to grow at a CAGR of around 15% during the period 2020-2025. Similarly, the plant-based milk market in India is also showing strong growth potential. The plant-based milk market in India is expected to grow at a CAGR of more than 20% between 2020 and 2025.

5.2 To study the consumer perception towards plant-based meat and milk products

The consumer perception towards plant-based meat and milk products in India is positive. While some consumers may still be unfamiliar with these products or sceptical of their taste and nutritional value, there is a growing awareness and acceptance of plant-based options in the Indian market. According to study, 69% of Respondents are aware of plant-based meat and milk alternatives and 32% of respondents are not aware of plant-based milk and meat products. Chi-square test was conducted to see association between awareness and demographic profiles and it is concluded that there is relationship between age and awareness. The study also found that Veganism and environmental concerns were the primary drivers of consumer interest in plant-based products.

5.3 To identify the factors influencing the growth of plant-based meat and milk products.

The Study revealed that Ingredients, taste and texture, price are important factors that influence consumer purchasing decisions. Consumers may be willing to try plant-based options, but they expect them to be comparable in taste and price to their animal-based counterparts. Others factors that do affect the purchase of plant-based meat and milk are

altruistic benefits and nutritional benefits as producing animal-based products requires large amounts of water, land, and other resources, and can contribute to deforestation, greenhouse gas emissions, and other environmental problems.

5.4 Recommendations

1. **Develop more locally-sourced ingredients:** India has a rich variety of plant-based ingredients that can be used to create meat and milk alternatives. By developing more locally-sourced ingredients, the industry can create unique products that cater to Indian tastes and preferences.
2. **Increase accessibility and affordability:** Plant-based meat and milk products are often more expensive than their animal-based counterparts, which can make them less accessible to the general population. The industry should focus on making their products more affordable, especially for lower-income consumers.
3. **Educate consumers:** Many consumers in India are not familiar with plant-based meat and milk products, and may not understand their benefits. The industry should focus on educating consumers about the health and environmental benefits of these products, and how they can be used in Indian cuisine.
4. **Innovate with traditional Indian recipes:** India has a rich culinary tradition, and plant-based meat and milk products can be used to create unique and innovative dishes that cater to Indian tastes. The industry should focus on innovating with traditional Indian recipes to create new and exciting products.
5. **Collaborate with local farmers:** Collaborating with local farmers can help the industry secure a steady supply of high-quality ingredients, while also supporting local communities. By working closely with farmers, the industry can create a sustainable supply chain that benefits everyone involved.
6. **Partner with food service providers and restaurants:** Partnering with food service providers and restaurants can help increase the availability and visibility of

REFERENCES

- ASSOCHAM. (2022). *National Conference plant-based food 2023*.
- GFI India & IPSOSIS. (2020). *INSIGHTS ON THE PLANT-BASED MILK IN INDIA*.
- Indian Retailer. (2022). <https://www.indianretailer.com/article/whats-hot/consumer-behavior/plant-based-meat-market-to-reach-rs-3-500-crore-by-2026.a8185>.
- institute, T. g. (2020). *Strategies to Accelerate Consumer adoption of plant-based meat*.
- Joel K. Kahn, M. (n.d.). *The Plabt-Based Solution*.
- MARKETSANDMARKETS. (2021-2026). *Plant-Based Meat Market in India: Industry Analysis, Trends, and Forecast*.
- MDPI. (2023). <https://www.mdpi.com/2071-1050/12/14/5662>.
- Mousel, X. T. (2016).
- Renub Research. (2020-2027). *Global Plant-Based Meat Market Report*.
- Science Direct. (2023). <https://www.sciencedirect.com/science/article/pii/S2214799322001217>.
- scispot. (2023). <https://www.scispot.com/blog/top-7-plant-based-meat-companies-in-india>.
- Statista. (2023). <https://www.statista.com/statistics/1362455/india-plant-based-food-market-size-by-segment/#:~:text=The%20estimated%20market%20size%20of,rupees%20during%20the%20same%20period>.
- Technavio. (2021-2026). *Plant-Based Milk Market - Global Outlook and Forecast*.
- USDA. (2021). *India Emerges as a Burgeoning Market for Plant-based Meat substitutes*.

APPENDICES

Google form

This form is regarding the awareness and preference of Plant-Based Meat and Dairy products. Kindly take few minutes to fill out the form.

* Required

1. Name *

2. Age *

Mark only one oval.

- ☐ <35 years
☐ 35 to 45 years
☐ >45 years

3. Location *

Mark only one oval.

- ☐ Delhi
☐ Bangalore
☐ Mumbai
☐ Chennai

4. Education Background *

Mark only one oval.

- ☐ No formal education
- ☐ Primary School
- ☐ Higher secondary school
- ☐ Graduate
- ☐ Post graduate and above.

5. Occupation *

Mark only one oval.

- ☐ Student
- ☐ Self-employed
- ☐ Government servant
- ☐ Others

6. Income (Annually) *

Mark only one oval.

- ☐ <3 lakhs
- ☐ 3-8 lakhs
- ☐ 8-30 lakhs
- ☐ >30 lakhs

7. Are you aware about plant-based meat and milk products *

Mark only one oval.

☐ Yes

☐ No

Plant-based Milk Products

8. What is the reason behind choosing plant-based products? *

Check all that apply.

☐ Economical

☐ Vegans can eat

☐ Animal welfare

☐ Health-conscious

☐ Taste

☐ Food curiosity

9. What is your monthly expenditure on plant-based products? *

Mark only one oval.

☐ <1000 Rs.

☐ 1000 Rs. to 2500 Rs.

☐ >2500 Rs. & Above

10. What is your source preference? *

Check all that apply.

☐ Cereal based products

☐ Legume based products

☐ Nut based products

☐ Seed based products

11. What is your product preference? *

Check all that apply.

- ☐ Soya Milk
- ☐ Almond Milk
- ☐ Cashew Milk
- ☐ Coconut Milk
- ☐ Hemp Milk
- ☐ Others (Quinoa, Oat, Rice, Pea, Flax)

12. What is your preferred distribution channel? *

Mark only one oval.

- ☐ Food Retail and Services
- ☐ E-commerce platform
- ☐ Other:

13. What is your Brand preference? *

Check all that apply.

- ☐ Goodmylk
- ☐ Urban Platter
- ☐ Raw Pressery
- ☐ Nutriva
- ☐ Sofit
- ☐ Other:

14. What factors influence the purchase of Plant-based Milk Products? *

Check all that apply.

- ☐ Price
- ☐ Taste
- ☐ Ingridients
- ☐ Packaging
- ☐ Previous experience
- ☐ Other:

Plant-based Meat Products

15. What is your source preference? *

Check all that apply.

- ☐ Soya based products
- ☐ Wheat based products
- ☐ Blends based products
- ☐ Pea based products
- ☐ Other Sources (quinoa, oats, beans, nuts, and seeds)

16. What is your product preference? *

Check all that apply.

- ☐ Strips and nuggets
- ☐ Burger patties
- ☐ Meatballs
- ☐ Sausages
- ☐ Others (Slices, cutlets, slides, fingers and crumbles)

17. What is your preferred distribution channel? *

Mark only one oval.

- ☐ Food Retail and Services
- ☐ E-commerce platform
- ☐ Other:

18. What is your Brand preference? *

Check all that apply.

- ☐ GoodDot
- ☐ Imagine Meats
- ☐ Wakao
- ☐ Evo Foods
- ☐ Vezlay
- ☐ Other:

19. What factors influence the purchase of Plant-based Meat Products?

Check all that apply.

- ☐ Availability of the produce
- ☐ Nutritional value
- ☐ Taste & Texture
- ☐ Accessibility of the produce
- ☐ Cost of the produce
- ☐ Free from additives
- ☐ Produced organically

